

Cost-benefit analysis

Quantifying valuation of non-marketed goods

Overview

- Revealed preference methods
 - Market analogy
 - Trade-off
 - Valuation
 - Hedonic price
 - Travel cost
- Stated preference methods
 - Direct elicitation
 - Dichotomous choice

Missing price for non-marketed goods

- **Change in social surplus** depends on number of affected individuals and marginal social benefit/cost.
- In ideal markets: $\text{price} = \text{marginal social cost} = \text{marginal social benefit}$.
- With distorted or missing markets, we seek a **shadow price**.
- **Non-marketed goods** (e.g., human life, leisure):
 - **Revealed preference**: infer shadow price from related markets using observed behavior.
 - **Stated preference**: contingent valuation via surveys; relies on reported willingness to pay/accept.
- Key distinction: revealed vs. stated preference → based on whether **actual behavior** is observed.

Revealed preference methods

- Value of non-marketed goods is inferred from **observed behavior** in related markets.
- Key subtypes:
 - **Market analogy** – use prices/expenditures of similar private-sector goods.
 - **Trade-off method** – value inferred from time/money spent on substitutes (e.g., water filters).
 - **Valuation method** – valuation based on capital good or intermediate goods
 - **Travel cost method** – estimate demand for recreational sites based on travel expenses.
 - **Hedonic pricing** – isolate value of attributes (e.g., environmental quality) from housing/wage differentials.

Market analogy method

- Governments often provide private goods (housing, campsites, education) at **below-market prices**.
- Price paid for these goods represents only a **single point on the demand curve**, not the full curve.
- By using analogous private-sector goods, we can estimate the true demand curve.

Using market prices/expenditures

- The value of a publicly provided good can be estimated using the **market price of a similar private good**.
- When the government charges a price lower than the market, the amount collected in fees or rent **underestimates the true benefit**, since many users would be willing to pay more.
- Example: A local government rents out 100 apartments at **5,000 kr/month** → revenue = **500,000 kr**.

But in the private market, similar apartments cost **10,000 kr/month**. If we use this as the shadow price, the benefit is closer to **1 million kr**.

- It's important to consider **who the users are**: If the apartments are occupied by wealthier households, their **willingness to pay (WTP)** is likely even higher, meaning the government's revenue figure captures only part of the benefit.

Using data to estimate demand

- Instead of only looking at the **average willingness to pay (WTP)**, it is better to think about the **demand curve** for a publicly provided good.
- Private-sector data can help us **map the demand curve** if a similar good exists.
- Just using expenditures underestimates benefits because it **ignores consumer surplus**.

Using data to estimate demand

- Example: A city offers **free public Wi-Fi**, used by **300,000 people daily** → this gives us one point on the demand curve (price = 0, users = 300,000).
- In another area, similar Wi-Fi costs **5 kr/day** and has **100,000 users** → another point on the demand curve.
- By connecting these points (assuming linear demand), we can estimate the whole demand curve and calculate **consumer surplus** (the extra benefit beyond what people pay).

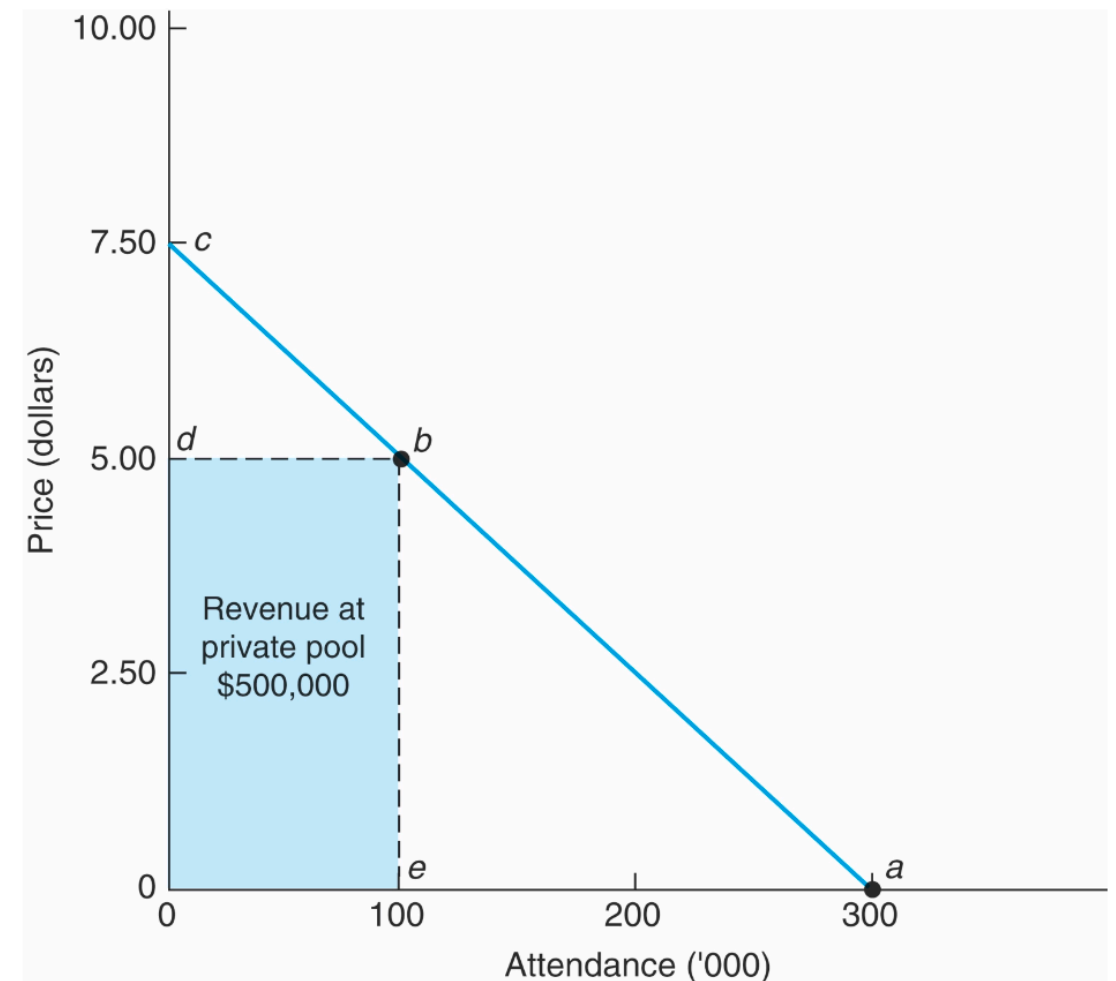


Figure 1: probable demand curve

Trade off method

- Concept: **Shadow prices** can be estimated from **opportunity costs** (what is given up).
- Examples:
- **Value of time saved** → after-tax wage as proxy.
- **Value of a statistical life (VSL)** → trade-offs between wages and fatality risk.

Valuing time saved

- Public projects may change **travel or waiting time**.
- Labor market provides a benchmark: $\text{wage} = \text{marginal value of time}$ (under ideal conditions).
- Challenges:
 - Wages exclude benefits (part of compensation).
 - Must use **after-tax wage** (plus benefits).
 - People value different types of time differently.
 - Market rigidities may prevent adjusting work hours.

Valuing a statistical life (VSL)

- Public projects may need to allocate resources for life-saving projects.
- Methods:
 - Forgone earnings** → discounted future income (problematic: undervalues women/elderly, may yield negative values).
 - WTP for safety devices** → e.g., paying 1,000 kr to reduce risk by 1/10,000 ⇒ $VSL = 10$ million kr.
 - WTP to avoid danger** → e.g., giving up 50,000 kr/year to reduce job fatality risk by 1/1,000 ⇒ $VSL = 50$ million kr.

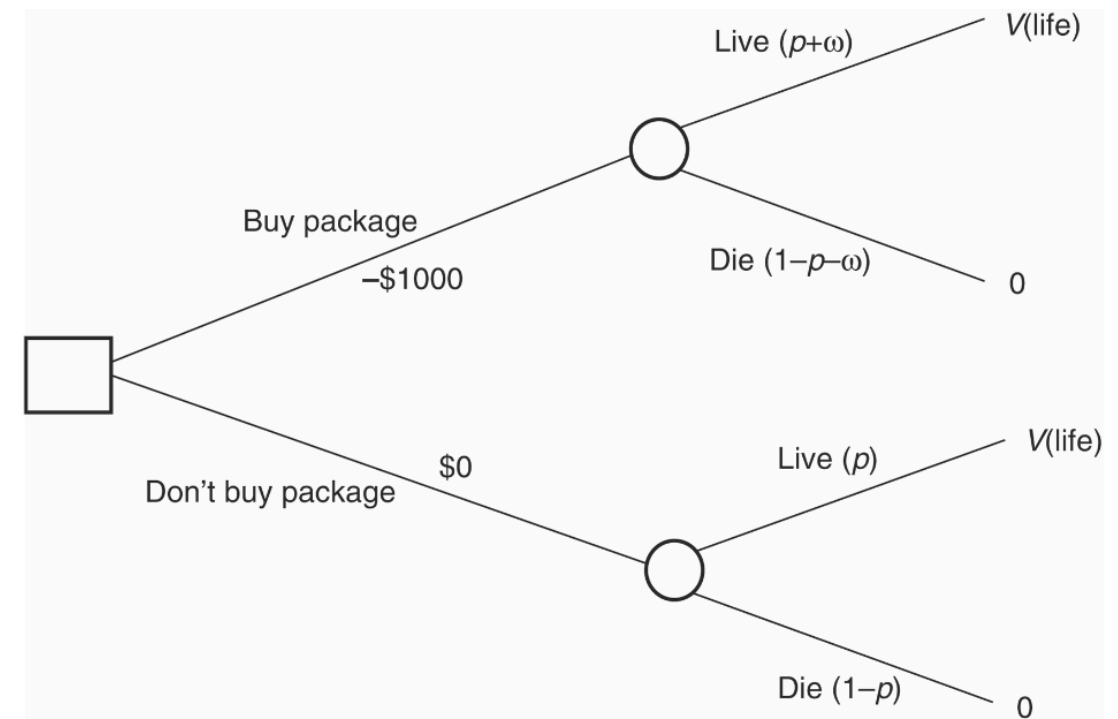


Figure 2: Decision tree for safety package purchase

Limitations of trade off methods

- There are several limitations to these simple approaches.
- Assumes people fully understand risks.
- Workers in risky jobs may not be **representative** of the general population.
- **Measurement of risk** may be inaccurate.
- WTP for safety depends on both **risk levels and changes in risk** → diminishing marginal utility of safety.

Valuation of related goods

- Using Capital Goods Prices:
 - Policy impacts can be inferred from **changes in capital goods prices**.
- Examples:
 - **Noise valuation** → compare house prices in noisy vs. quiet neighborhoods.
 - **Regulatory changes** → shifts in firm market value = change in producer surplus.
- **Advantage:** markets quickly capitalize information, so **price differences reflect value** of policy changes.

Valuation of related goods

- Using Non-Traded Intermediate Goods:
 - If project output is not traded, its value is inferred from **downstream impacts**.
- Example: irrigation project → value estimated from changes in **farmers' net income (NI)**.

$$\text{Annual Benefit} = \text{NI}_{\text{with}} - \text{NI}_{\text{without}}$$

- Total benefit = discounted sum of annual benefits over project life.
- Applications: irrigation, human capital (training programs → compare incomes of participants vs. non-participants).
- **Limitation:** assumes income difference captures all benefits → may miss extra consumption benefits.

Valuation of related goods

- Critique of Simple Valuation Methods:
- **Omitted variable problem:** policy may affect more than the observed good/market.
- **Self-selection bias:** participants in a program/project may differ systematically from non-participants.
- These issues can bias estimates of benefits from capital goods or intermediate goods.

Hedonic price method

- Concept: Values **attributes of goods** (e.g., scenic view, pollution, job risk) reflected in prices of **houses** or **wages**.
- Conducted in **two steps**:
 - Estimate **hedonic price function** (impact of attributes on price).
 - Derive **Willingness to Pay (WTP)** for the attribute, controlling for tastes/income.
- Helps address omitted variable and self-selection bias issues.

Hedonic price method

- Step 1: Hedonic Price Function

- Regression:

$$\ln P = b_0 + b_1 \ln(\text{area}) + b_2 \ln(\text{view}) + b_3(\text{close to city}) + b_4(\text{other char.}) + \varepsilon$$

- Coefficient b_2 : elasticity of house price w.r.t. scenic view.
Hedonic (implicit) price of view:

$$h_v = \frac{\partial P}{\partial \text{view}} = b_2 \cdot \frac{P}{\text{view}}$$

- Interpretation: extra cost of a small improvement in scenic view.

Hedonic price method

- Step 2: WTP for Attributes
- We can estimate the following WTP function (inverse demand function) for scenic views:

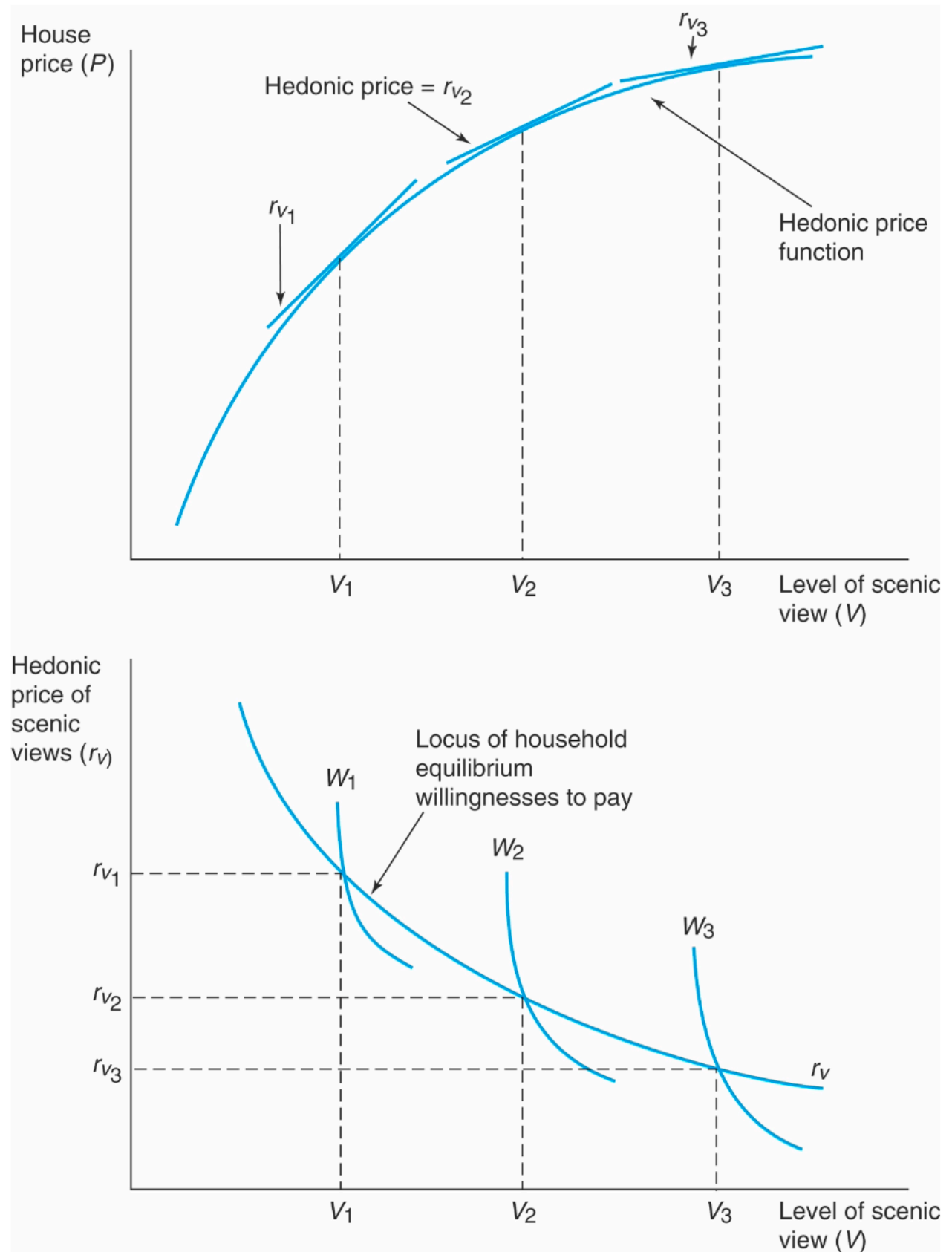
$$h_v = W(\text{view}, Y, Z)$$

h_v : implicit price of view,

Y : income,

Z : household characteristics (reflecting tastes).

- From this, we can compute **changes in consumer surplus** for improvements in scenic view.



Hedonic price method

- Limitations of Hedonic Models
 - Assumes people understand the attribute (e.g., pollution levels and health risks).
 - **Econometric issues:** measurement error, specification error.
 - **Market frictions:** prices may not fully adjust in the short run.

Travel cost method

- Used mainly to estimate the value of **recreational sites**.
- Visits depend on:
 - True cost of visiting; Price of substitutes; Income levels; Other preference-related factors.
- Total cost of visit includes:
 - Admission fee; Travel costs (time, transport, fuel); Accommodation, parking, etc.

Travel cost method

- How TCM Works:
- Admission fees often don't vary (or are zero), but travel costs differ across individuals.
- Variation in travel costs → variation in usage patterns → allows estimation of a **demand schedule**.
- Steps:
 - Sample households around the site.
 - Survey: number of visits, total travel cost, substitutes, income, other characteristics.
 - Specify and estimate a demand function.

Travel cost method

- Limitations of TCM:
 - **Measurement error** in costs and visits.
 - **Selection bias / endogeneity**: frequent visitors may differ systematically from others.
 - **Omitted variable bias**: unobserved factors influencing visits not accounted for.

Stated preference method

- Used when **preferences cannot be observed** from behavior (e.g., public goods).
- Relies on surveys → commonly called *contingent valuation surveys* (CVS).
- Aim: elicit **willingness to pay (WTP)** for changes in quantity/quality of a good.
- Non-controversial for **direct-use goods** (e.g., parks, recreation).
- More controversial for **non-use goods** (e.g., preserving archaeological sites).
- Use of CVS is **rapidly growing**.

Direct elicitation method

- Open-Ended: respondents state their maximum WTP directly.
- Close-Ended Iterative Bidding: respondents accept/reject a price; amount adjusted up/down until WTP found.
- Limitation: very sensitive to the starting value.
 - Contingent Ranking: respondents rank bundles of goods and payments (e.g., water quality vs. taxes).
 - Easier for respondents, but WTP must be inferred from rankings.
 - Results can be influenced by the order of alternatives.

Dichotomous Choice (Survey Valuation Method)

- Respondents asked if they would pay a **specific bid price** for a good/policy.
- With many respondents:
 - Accept/reject probabilities calculated for each bid.
 - Plot histogram → fit curve = **demand curve** for a typical individual.
 - Area under curve** = estimate of WTP.
- Challenges:
 - Defining the good/policy clearly (esp. for **non-use goods**).
 - Non-commitment bias** → respondents don't actually pay.
 - Iterative bidding** sensitive to starting value/order.
 - Judgmental biases**: WTP vs. WTA wording → often $WTA > WTP$ (linked to **loss aversion**).

Reading materials

- Boardman et al. Chapter 15.
- Veileder i samfunnsøkonomiske analyser, section 3.4.

Cost-benefit analysis

Discounting future impacts

Overview

- Basics of discounting
 - Net present value, internal rate of return
- Nominal vs real return
- Project selection
- Selection of discount rate
 - Current practices in Norway

Basics of discounting

- Goal: convert **future values** into today's values → **Present Value (PV)**.
- Example: Invest X today, interest rate = i .
 - Future value after 1 year: $Y = X(1 + i)$.
 - Present value of Y :

$$PV = \frac{Y}{1 + i} = X$$

- Interpretation: PV is the amount today that is **equivalent to a future payoff**.

Basics of discounting

- **Extending to multiple years**

- For Y received in year T :

$$PV = \frac{Y}{(1 + i)^T}$$

- Discount factor:

$$\delta_T = \frac{1}{(1 + i)^T}$$

- This is **exponential discounting** → value falls more the further into the future.

Basics of discounting

- **Streams of benefits and costs**

- Projects usually have **flows over many years**.
- Present value of benefits:

$$PV(B) = \sum_{t=0}^T \frac{B_t}{(1+i)^t}$$

- Present value of costs:

$$PV(C) = \sum_{t=0}^T \frac{C_t}{(1+i)^t}$$

- **Net Present Value (NPV) = $PV(B) - PV(C)$.**
- Decision rule:
 - If **NPV > 0**, project adds value → accept.
 - If **NPV < 0**, reject.

Basics of discounting

- Effect of interest rate:
Higher *interest rates* → future benefits valued less → **NPV decreases**.
- With zero interest rate:
NPV = simple sum of annual net benefits.
- Example:
Total benefits (no discounting): **750,000**.
Total costs (no discounting): **625,000**.
Net = **125,000**.
- With larger interest rate, NPV falls below 125,000 because future benefits shrink in present terms.

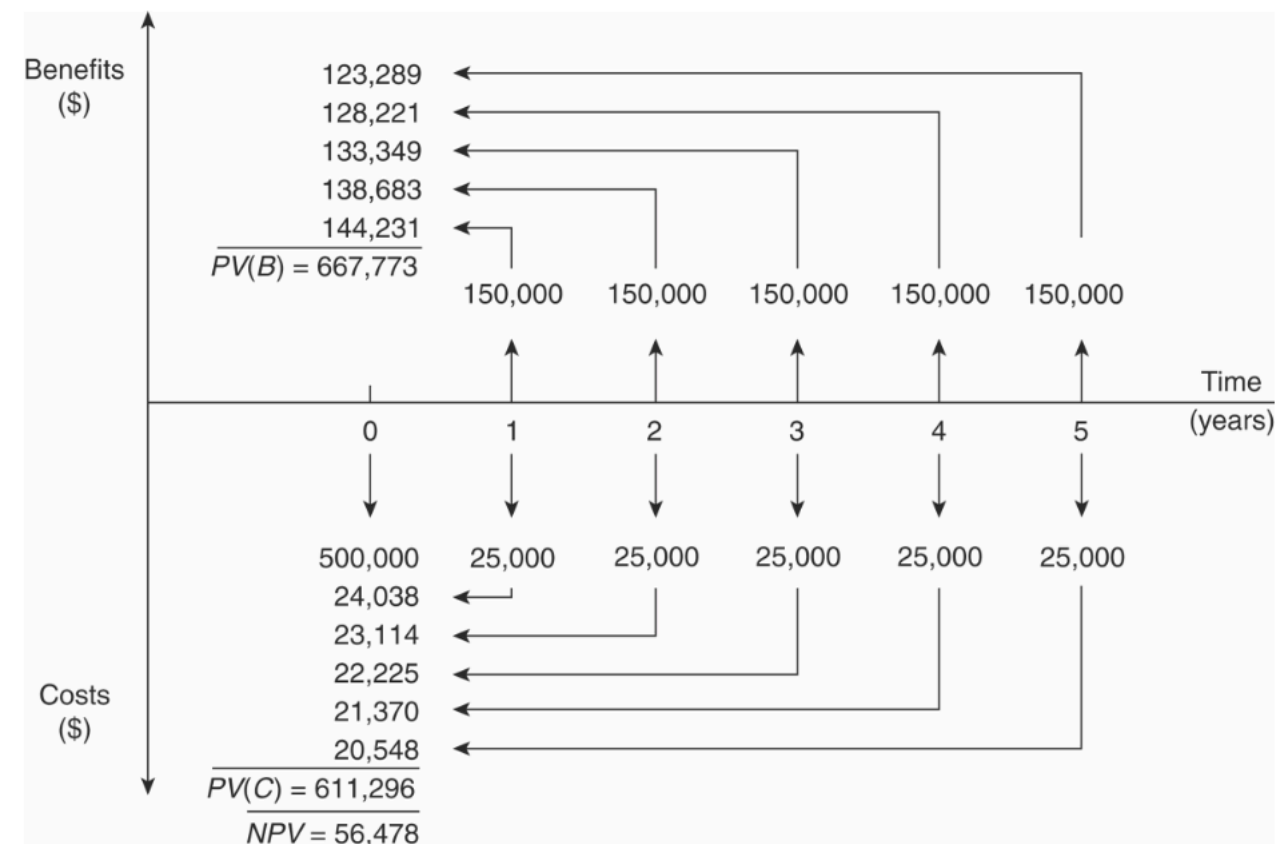


Figure 6: A project with streams of future benefits and costs

Internal rate of return

- The Internal Rate of Return (IRR) of a project is the interest rate at which the project's NPV equals zero.
- Sometimes, IRR provides a good indication of the desirability of a project.
- IRR is the highest interest rate that leaves the project profitable.

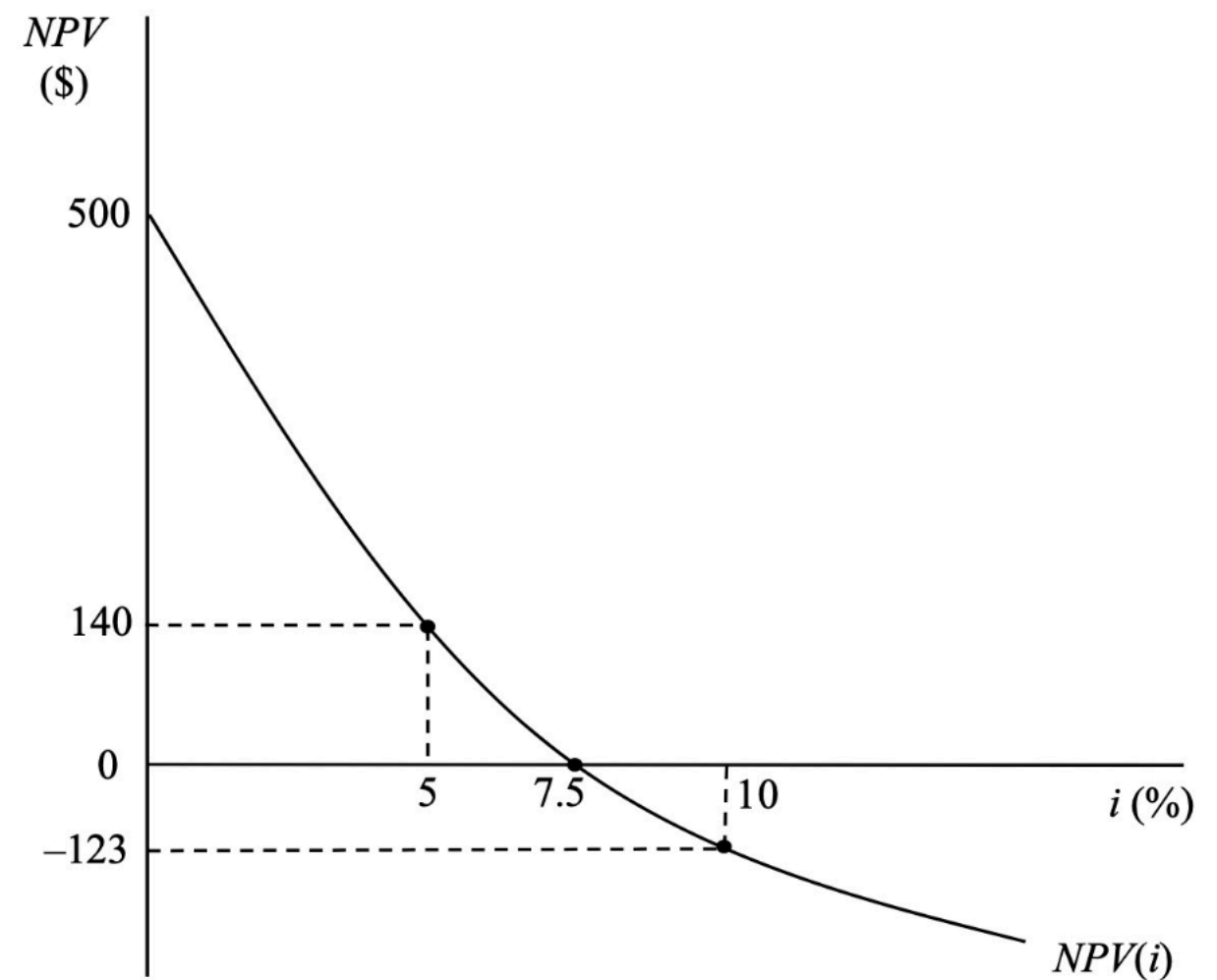


Figure 7: Internal rate of return

Compounding over multiple sub-periods

- So far, we assumed **annual periods** and an **annual interest rate**.
- But costs/benefits may occur **more frequently** (e.g., semi-annual, quarterly). Then we need to adjust the interest rate.

- Formula:

$$(1 + i_s)^s = (1 + i)$$

- Rearranging,

$$i_s = (1 + i)^{\frac{1}{s}} - 1$$

- Example:

Annual rate $i = 10\%$.

Quarterly compounding ($s = 4$):

$$i_s = (1.10)^{1/4} - 1 \approx 2.41\% \text{ per quarter.}$$

Some useful formulae

- When returns are **constant each period**, we can use the formula for a **geometric series**.

General geometric sum (finite):

$$a + ad + ad^2 + \dots + ad^n = a \cdot \frac{1 - d^{n+1}}{1 - d}, \quad d \neq 1$$

Infinite series (as $n \rightarrow \infty$):

$$a + ad + ad^2 + \dots = \frac{a}{1 - d}, \quad 0 < d < 1$$

- If a project pays a constant benefit V each year:

$$\sum_{t=0}^T \frac{V}{(1+i)^t} = V \cdot \frac{1+i - (1+i)^{-T}}{i}$$

- If payments start from year 1 instead of year 0:

$$\sum_{t=1}^T \frac{V}{(1+i)^t} = V \cdot \frac{1 - (1+i)^{-T}}{i}$$

- Difference: whether benefits begin **immediately (t=0)** or **after one period (t=1)**.

Annuity and perpetuity

- **Annuity**: a fixed payment V each year for T years.
- **Perpetuity**: an annuity that goes on **forever**.
- Common in Cost-Benefit Analysis (CBA).

Annuity (payments start at year 1):

$$PV = V \cdot \frac{1 - (1 + i)^{-T}}{i}$$

Perpetuity:

$$PV = \frac{V}{i}$$

- Simple shortcut: perpetual payments are valued as payment divided by interest rate.

Equivalent annual net benefit (EANB)

- Sometimes we want to convert NPV of a project in terms of an annuity.

Define V such that:

$$NPV = V \cdot \frac{1 - (1 + i)^{-T}}{i}$$

Then:

$$EANB = \frac{NPV}{\text{annuity factor}}$$

Annuity factor:

$$\frac{1 - (1 + i)^{-T}}{i}$$

An example

Suppose: receive 1,000 kr/year for 5 years at $i = 10\%$.

Step 1: Annuity factor

$$\frac{1 - (1.1)^{-5}}{0.1} \approx 3.791$$

Step 2: PV of annuity

$$PV = 1,000 \times 3.791 \approx 3,791 \text{ kr}$$

Step 3: If NPV = 3,791 kr, then

$$EANB = \frac{3,791}{3.791} = 1,000 \text{ kr/year}$$

Nominal vs real returns

- Question: Should future benefits/costs be measured in **nominal** (with inflation) or **real** (without inflation) terms?
- Rule: Be **consistent** – use either:
 - **Nominal values** + nominal discount rate, or
 - **Real values** + real discount rate.
- Both approaches give the **same decision** if applied consistently.

Nominal vs real returns

- Formula:

$$(1 + r)(1 + m) = (1 + i)$$

- i = nominal interest rate.
- m = inflation rate.
- r = real interest rate.
- Solving for r :

$$r = \frac{i - m}{1 + m}$$

- Advantage of real terms:
 - More intuitive (inflation-adjusted).
 - Allows flexible use of **different inflation forecasts** for future periods.

Project selection

- Why NPV?
- After discounting, we need a **decision rule** for projects.
- Net Present Value (NPV): key criterion.
- Rule for a single project:
Accept if **NPV > 0**.
Reject if **NPV < 0**.
- Intuition:
Positive NPV = benefits exceed opportunity cost.
Negative NPV = funds better used elsewhere (status quo).

Project selection

- NPV with Multiple Projects:
- If comparing mutually exclusive projects:

Choose the project with the **highest NPV**, provided NPV is positive.

If all NPVs are negative → stick with the **status quo**.

- NPV measures **net value in monetary terms**, which makes comparisons across projects straightforward.

Project selection

- Another indicator for assessment is IRR (Recall that IRR is the Interest rate that makes $NPV = 0$)
- Rule: Accept if $IRR > \text{discount rate}$.
- Between two projects, may prefer the one with the **higher IRR**.
- But there are problems with IRR-based rules:
 - May not be **unique** (multiple IRRs possible).
 - Expressed as a **percentage**, not monetary value.
 - Not suitable when comparing projects of **different sizes**.

Project selection

- How to compare projects with different lifespan? Direct NPV comparison is misleading if projects have **different time frames**.
- Two solutions:
 - Roll-over method:
Repeat the shorter project until its time matches the longer project.
Example: A = 30 years, B = 15 years → compare NPV(A) vs. NPV(2 × B).
Formula:

$$NPV = x + \frac{x}{(1+i)^{15}}$$

where x is NPV of one 15-year project B.

—Converts NPV into an annual equivalent payment.

$$EANB = \frac{NPV}{\text{annuity factor}}$$

Gives per-year comparability across projects.

Limitation: Does not fully account for **different project lifespans**.

Selection of discount rate

- Why Do We Need a Social Discount Rate (SDR)?
- A **discount rate** determines how we weigh **future benefits and costs** relative to today.
- Choosing a discount rate = deciding the **weights applied to each year's impacts**.
- For government projects, this becomes a **social choice** → the **Social Discount Rate (SDR)**.
- Two broad approaches to determine SDR:
 - Rate of Time Preference method.
 - Opportunity Cost of Capital method.

Selection of discount rate

- **Individual time preference**
- Individuals compare consumption **today vs. in the future.**
- Marginal Rate of Time Preference (MRTP):
 - It reflects an individual's willingness to trade off future for present
 - If we could define a general time-contingent utility function, then MRTP measures the slope of the indifference curve that allows individuals to substitute utility across time periods

Individual time preference

- **M RTP and the Market Interest Rate**
- In a perfect capital market (no frictions):
 - People can borrow and lend freely.
 - Each individual adjusts until:

M RTP = Market interest rate

- Key implication:
 - Even if individuals differ in tastes, their **observed MRTPs align with the interest rate**.
 - Thus, the market interest rate can be taken as the social rate of time preference.

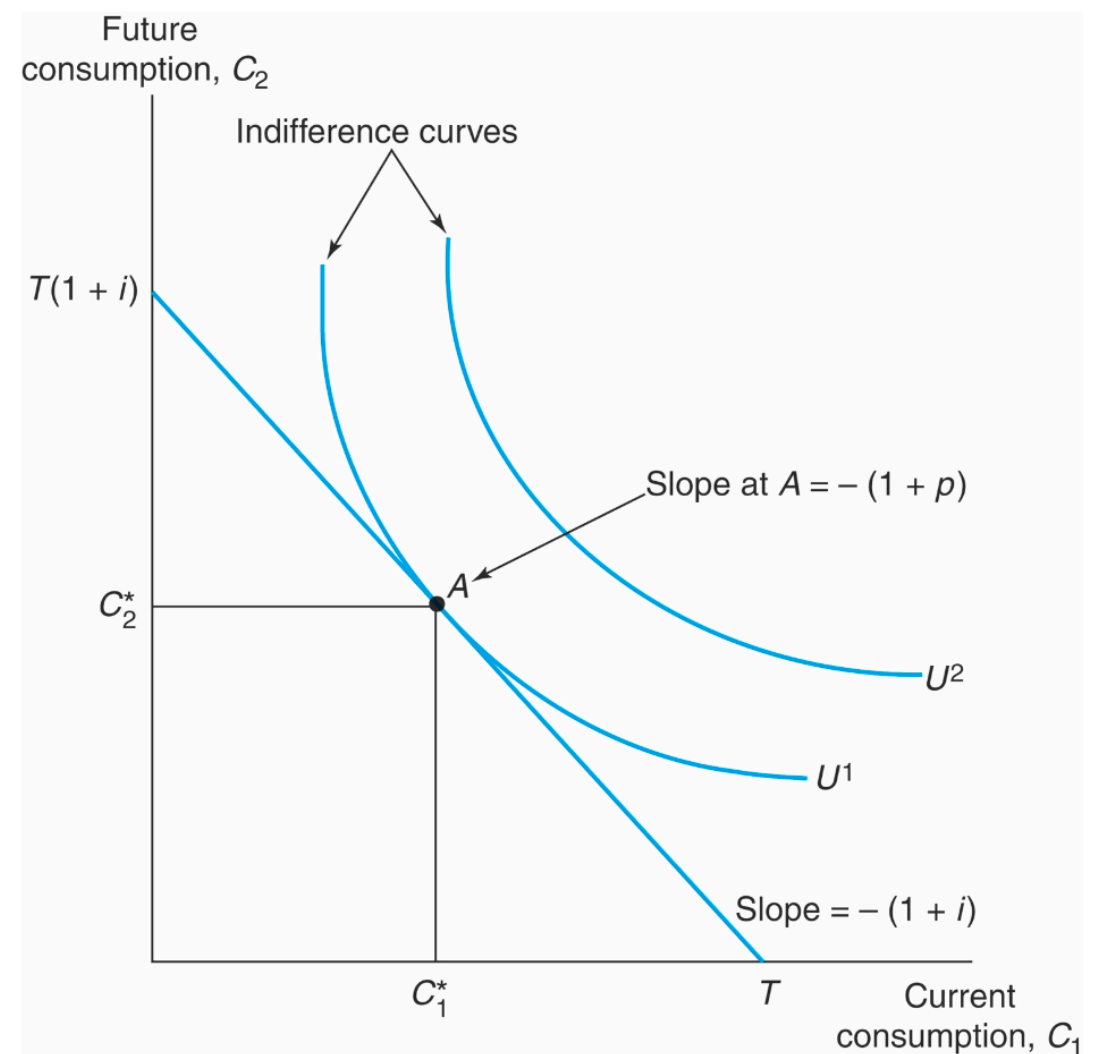


Figure 9: MRTP equals the market interest rate

Individual time preference

- **Extending to Production (ROI Connection)**
- In a closed economy with production: If no market failures, taxes, or transaction costs

$$MRTP = i = ROI$$

- Implying: The rate of time preference, the interest rate, and the marginal return on investment all must coincide.
- In this “perfect” setting: A single common interest rate emerges.
- This is the natural candidate for the **Social Discount Rate (SDR)**.

Social opportunity cost of capital

- Origin: **Harberger's work** on closed domestic markets.
- Idea: a new government project is funded by **borrowing** → raises interest rate.
 - Results in less private investment and more savings.
- SDR should reflect the **social cost** of diverting resources:
 - Weighted average of—Marginal return on private investment, and After-tax marginal return on savings (foregone consumption).

Ramsey's social time preference method

- Developed by **Frank Ramsey**.
- Society maximizes a social welfare function over time.
- SDR = society's marginal rate of time preference (STP):

$$STP = \rho + \mu g$$

where

rho is pure rate of time preference (impatience),
g is growth rate of per capita consumption,
mu is elasticity of marginal utility of consumption.

- Interpretation:
rho measures how society values present consumption more than future,
where as mu times g together trades off the possibility of richer future generations with potential lower marginal value of extra consumption.

Issues with intergenerational projects

- Constant SDR may be fine for **same generation projects**. But problematic for **long-term impacts** (climate, health, environment).
- Issues with constant SDR:
 - Future generations' preferences ignored.
 - Distant benefits almost vanish (tiny PVs).
 - Ignores uncertainty in future market rates.
- One way to address it is to use time-declining discount rates. There are empirical evidence that people use **lower rates** for distant future events.
- Policy recommendation:
 - Use a declining discount rate schedule.
 - Gives more weight to **long-term benefits/costs**.
 - Helps make **intergenerational projects** (e.g., climate policy) look more cost-beneficial.

Historical guidelines (1967–1999)

- 1967 Johansen Report: Based on Ramsey model.
—Assumptions: $\rho = 1\%$, $\mu = 3$, $g = 3\%$ → SDR = **10%** (Circular R-3/1975).
- 1978 (R-25/78): lowered to 7%.
- 1997 NOU Green Paper: introduced CAPM approach.
- 1999 Circular R-14/99:
 - Risk-free real rate: **3.5%**.
 - 3 risk classes: **4%, 6%, 8%**.
 - Large projects → specific risk-adjusted rates recommended.

Current practice

- Circular R-109/2005:
 - Risk-free rate = **2%**.
Normal project: risk premium = **+2%** → SDR = **4%**.
High systematic risk: premium = **+4%** → SDR = **6%**.
 - Large/complex projects: specific rate calculations required (esp. quality-assured projects).
- General Principle (NOU 2012:16)—The real risk-adjusted SDR should reflect the risk-free interest rate, and the risk of the project (risk-adjusted opportunity cost of capital).
- Sector-Specific Practice:
 - Ministry of Transport & Communications (post-2005): Applied 4.5% discount rate across road, rail, aviation.
 - Chosen for **practical uniformity**, not differentiated by project type.

Key takeaways

- Shift over time:

Ramsey model → CAPM → risk-adjusted, declining long-term rates.

- Current standard:

4% (0–40 yrs), 3% (40–75 yrs), 2% (75+ yrs).

- Flexibility remains:

Projects with higher risk → higher SDR.

Large/unique projects → tailor-made rates.

Reading materials

- Boardman et al. Chapters 9 and 10.
- Veileder i samfunnsøkonomiske analyser, section 3.5.